

REAL EXCHANGE RATE DYNAMICS WITH NOMINAL PARITIES

Structural Change and Overshooting

Guillermo A. CALVO*

Department of Economics, University of Pennsylvania, Pennsylvania, PA 19104, USA

Received June 1984, revised version received April 1986

The relationship between structural changes that require a modification of the steady-state level of the real exchange rate is examined in terms of a model of staggered prices and real-wealth effects. Although wealth effects are necessary for overshooting, it is shown that their actual values are of no particular relevance; in fact, overshooting is shown to be more likely the smaller is the real-wealth effect in the demand for tradables. A key (and counterintuitive) result is that overshooting is more likely the higher is the frequency of price changes.

1. Introduction

The recent stabilization experience of the Southern Cone of Latin America has spawned a great deal of interest in the theory of fixed or, more generally, predetermined exchange rate systems. The main focus of this kind of research has been the connection between a slowdown in the rate of depreciation of the *nominal* exchange rate and a protracted appreciation of the *real* exchange rate [see, for example, Dornbusch (1982), Rodriguez (1982), Calvo (1982a, 1983) and Obstfeld (1984)].

Despite its theoretical interest, however, the above-mentioned exchange rate policy was only one of a variety of other measures that went simultaneously into effect; prominent among this group were banking reforms and trade liberalizations of different degrees [see, for example, Rodriguez and Sjaastad (1979), Diaz-Alejandro (1981), Harberger (1982), Calvo (1983, 1985a) and Edwards (1985)]. Therefore, the question arises: Is there a sound reason for expecting that the above-mentioned 'structural' changes were partly responsible for the protracted appreciation of the real exchange rate?

*This is an abbreviated and substantially revised version of Calvo (1982c), which was presented as the Miguel Sidrauski Lecture at the III Latinamerican Meetings of the Econometric Society, Mexico City, Mexico, August 1982. I am very thankful to two anonymous referees for useful comments. This work was partly financed by a grant from the National Science Foundation.

The link between structural changes and overshooting is really easy to establish in models with ad hoc expectations, even when those changes are assumed to be permanent. However, this situation is radically modified when expectations are assumed to be formed in a rational way. In simple models it is not unusual to find that once-and-for-all structural changes induce a monotonic behavior of the equilibrium real exchange rate [see Calvo (1982b) and Obstfeld (1984)], in which case overshooting would not occur. The search for the 'minimal' model where structural changes gives rise to nonmonotonic behavior is, therefore, a challenging, and possibly illuminating, pursuit.

Our purpose is to contribute to the understanding of the relationship between structural changes and overshooting of the real exchange rate in an environment of rational expectations. The central argument of the paper is that the combination of the assumption that prices are set in a non-synchronous manner and a Pigou effect go a long way in explaining overshooting, and that, much more interestingly, overshooting is more likely to occur the shorter is the average period of a price quotation. Since one would expect that the average 'life' of a price quotation tends to be relatively short in 'high' inflation economies – like those in the Southern Cone before the above-mentioned stabilization policies – the above result suggests that the overshooting phenomenon is more likely the higher is the initial inflation rate.

The basic model borrows heavily from Calvo (1982a); we will, therefore, assume the existence of tradable and nontradable goods, and the price of tradable goods in terms of foreign currency is exogenously given. Hence, assuming away commercial barriers, their price in domestic currency is simply the product of the international price by the nominal exchange rate (i.e. the price of foreign in terms of domestic currency).

The situation in the nontradable or home goods market is considerably more complicated. We assume that the economy is populated by a large number of outlets offering home goods, each one of which is forced to quote its price in terms of domestic currency for a period, the length of which depends on a stochastic signal (the 'green light'). We assume that individuals are aware of the structure of the model, and that, in their price-setting formula, they introduce future information which is presumably relevant for profit maximization. As shown in Calvo (1982a), if one assumes that aggregate demands are a function of relative prices and the rate of interest, it is possible to show, by adding some strategic mathematical assumptions – a key one being that the green light is stochastically independent across individuals, and that it has the same geometric distribution for all the individuals – that the system boils down to two ordinary differential equations with one 'state' variable (the average price of home goods), and one 'control' variable (the price set by those outlets which get the green

light). This simplicity engenders two strong results: (1) if the system is at a steady state, a once-and-for-all change in the rate of devaluation has no real effect, the rate of inflation simply adjusts to a new level; and (2) the approach to the steady state, given constant policy, is monotonic.

Result (1) implies that, starting at the steady state, this economy's rate of inflation can be immediately changed to a new permanent level by announcing a new permanent rate of devaluation: no painful adjustment process will be witnessed even when prices are, no doubt, 'sticky'. This is an interesting implication because it shows that models that are able to exhibit real effects associated with changes in the rate of devaluation are by no means belaboring the obvious. Furthermore, this exchange rate 'neutrality' is a feature that we wish to keep in the present paper, because it will assure us that we are getting insights that are independent of those obtained by the above-mentioned literature that emphasizes the real effects of once-and-for-all changes in the rate of devaluation.

In this paper we define a 'structural change' as any change that gives rise to a new steady-state level of the real exchange rate (defined as the ratio of the domestic price of tradables to the average price of nontradables). (In addition, and in order to shy away from presenting a boring taxonomy, we restrict our attention to cases where the real exchange rate is the *only* variable which the steady-state value affects.) Hence, result (2) above implies that if, say, a structural change calls for a fall in the steady-state real exchange rate, the approach to the new level will be monotonic, there will not be overshooting. This is, obviously, an implication that will have to go away if we are going to fulfill our promise.

From a formal point of view the main addition to Calvo (1982a) is an asset or Pigou effect in the aggregate demand functions. We do it in such a way that result (1) above remains intact.

Asset effects are obviously not new in this literature. They play a prominent role in Kouri (1976) and Calvo and Rodriguez (1977), for example, and their micro foundations are quite trivial. However, there is a general feeling of dissatisfaction when a given argument is strongly dependent on the existence of a quantitatively significant asset effects, mostly because empirical estimates suggest that they are relatively small. In this respect the present model is particularly satisfactory, because we are going to be able to show that *overshooting is more likely to occur the smaller is the asset effect on the demand for tradable goods*. Furthermore, although it is necessary for overshooting that the asset effect in the aggregate demand function for nontradables be nonzero, its actual value is of no particular significance for the existence of overshooting.

The paper is organized as follows. Section 2 presents the basic model. Section 3 derives the central results, while section 4 discusses some extensions. Proofs that do not directly help the economic intuition are relegated to the appendix.

2. The basic model

This is a two-sector economy: tradables and nontradables (or home goods). There are no commercial barriers and the country is a price-taker in the international market for tradables; thus, without loss of generality, we assume that the price of tradables in terms of foreign currency is identically equal to unity. Furthermore, since, as stated in the Introduction, we are not interested in the monetary experiment of reducing the rate of devaluation, we will assume that the exchange rate is constant over time and also equal to unity.

Pricing in the 'home' sector is subject to the following price-setting technology. We assume that each individual firm can quote its price only in domestic currency, and that it can change it only at the time a 'price-change' signal is received (the 'green light'). The probability (density) of receiving such a signal s periods from today is assumed to be independent of the last time the firm got it (i.e. the signal) and to satisfy

$$\delta e^{-\delta s}, \quad \delta > 0. \quad (1)$$

Therefore the average length of a price quotation is simply $(1/\delta)$.

For the sake of simplicity we assume that firms or outlets have the same price-setting equation. Let v_t denote the (log of the) price set at time t by a firm which gets the green light at time t , and p_t denote the (log of the) geometric price level for home goods at time t . Following Calvo (1982a) we assume

$$v_t = \delta \int_t^{\infty} [p_s + \beta f(p_s, a_s)] e^{-\delta(s-t)} ds, \quad \beta > 0, \quad (2)$$

where a_s is 'wealth' at time s in terms of tradables, and $f(\cdot, \cdot)$ is a measure of excess demand in the home-goods market.

On the other hand, p_t is assumed to satisfy

$$p_t = \delta \int_{-\infty}^t v_s e^{-\delta(t-s)} ds. \quad (3)$$

One way to justify (3) is to envision the home-goods sector as composed by a continuum of firms, and the price-change signal as being stochastically independent across firms. Thus, if we normalize the total number of firms to unity, we can appeal to the 'law of large numbers' to deduce, from (1), that δ firms will receive the signal per unit of time. Moreover, of the firms that last set their price at time s , a proportion,

$$e^{-\delta(t-s)}, \quad (4)$$

will not have received the price-change signal at time t ($> s$). Consequently,

$$\delta e^{-\delta(t-s)} \quad (5)$$

is the 'number' (more precisely, the 'measure') of firms which last set their prices at time s , and have not yet received a price-change signal at time t . Thus, if we define p as the arithmetic average of the v 's, weighted by the number of firms with the same v , (3) follows.¹

Granted existence, we can now differentiate (2) and (3) with respect to time to get:

$$\dot{v}_t = \delta[v_t - p_t - \beta f(p_t, a_t)], \quad (6a)$$

$$\dot{p}_t = \delta[v_t - p_t]. \quad (6b)$$

In order to keep the model within tractable limits, we are going to assume that there is no capital accumulation and that foreign assets pay no interest [as in Calvo and Rodriguez (1977)], in which case

$$\dot{a}_t = \text{trade balance at } t. \quad (7)$$

Thus, indicating by y^T the output of tradable goods (assumed constant)² and by $c^T(p, a)$ the absorption-of-tradable-goods function, we can write (7) as follows:

$$\dot{a}_t = y^T - c^T(p_t, a_t) \equiv -z(p_t, a_t), \quad (8)$$

where, obviously, $z(\cdot, \cdot)$ is the trade-balance-deficit function.

An equilibrium path (from $t=0$) is defined as a set of continuous paths (from $t=0$) for v , p and a that converge to their corresponding steady-state levels.³ Clearly, p and a are 'predetermined' at $t=0$ because the price level at $t=0$ is given by previous price quotations, and the level of assets can only change by saving or dissaving.

We will further restrict the basic model to satisfy the following conventional-wisdom type assumptions:

$$f_p < 0; \quad f_a > 0, \quad (9a, b)$$

$$z_p > 0; \quad z_a > 0. \quad (9c, d)$$

¹In Calvo (1982b), (3) is derived by starting from a discrete-time version of the model.

²This assumption saves us from the thorny task of specifying a supply function of tradables when there is a spectrum of disequilibrium prices in the home-goods sector.

³This definition is in line with Sargent and Wallace (1973) and Calvo (1977).

In words, (9) states that an appreciation of the real exchange rate (i.e. a rise in p) will reduce excess demand in the home goods market, (9a), and deteriorate the balance of trade, (9c), while an increase in wealth deteriorates the trade balance, (9d), and stimulates excess demand in the home-goods market, (9b). In section 4 the implications of relaxing (9a) – and, thus, accounting for the ‘contractionary devaluation’ case – are going to be briefly discussed.

Before we get on with the analysis, a brief comparison with related models is in order. In the first place, note that if prices are assumed to be fully flexible, $f(p, a) = 0$ for all t , and, hence, in equilibrium, by (9a, b):

$$p = \psi(a), \quad \psi'(a) > 0. \quad (10)$$

Thus, by (8) and (10):

$$\dot{a}_t = -z(\psi(a_t), a_t). \quad (11)$$

By (9c, d) and (10), the r.h.s. of (11) is a decreasing function of assets, a ; therefore, solutions would *monotonically* converge to the steady state. As can be easily verified, this is the fixed-rates versions of Calvo and Rodriguez (1977), and, as anticipated in the Introduction, does not yield the type of nonmonotonicity required for overshooting.

On the other hand, if the assets effect is eliminated from the excess-demand-for-home-goods function, i.e. if $f_a \equiv 0$, then the analysis of the real exchange rate could be simply performed in terms of (6). Granting uniqueness, and recalling that p would in this case be the only predetermined variable, there must be some function $\xi(\cdot)$ such that

$$v = \xi(p) \quad (12)$$

along the equilibrium path. Thus, by (6b) and (12),

$$\dot{p}_t = \delta[\xi(p_t) - p_t] \quad (13)$$

along the equilibrium path. Since the notion of equilibrium requires stability of (13), equilibrium p will, again, monotonically converge to its steady state. This is essentially the model studied in Calvo (1982a).

Consequently, as stated in the Introduction, the present model constitutes one of the simplest extensions of previous models that, as will be argued below, could account for some interesting cases of real exchange rate overshooting.

3. The workings of the model

We will examine the linear version of (6) and (8) which is obtained by linearization at the steady state. Without loss of generality we assume that the origin is the steady state of the system. Thus, we get the following first-order approximation for (6) and (7) at the origin:

$$\begin{pmatrix} \dot{v} \\ \dot{p} \\ \dot{a} \end{pmatrix} = B \begin{pmatrix} v \\ p \\ a \end{pmatrix}, \quad (14)$$

where

$$B = \begin{pmatrix} \delta & -\delta(1 + \beta f_p) & -\delta \beta f_a \\ \delta & -\delta & 0 \\ 0 & -z_p & -z_a \end{pmatrix}, \quad (15)$$

and where derivatives are evaluated at the steady state. Thus,

$$\text{trace of } B = -z_a < 0, \quad (16a)$$

$$\text{determinant of } B = \delta^2 \beta (z_p f_a - z_a f_p) > 0. \quad (16b)$$

Hence, matrix B has one positive characteristic root, while the other two have negative real parts. This is essentially what we need to ensure that given the initial conditions on p and a (p_0 and a_0), there exists a unique convergent continuous path satisfying (6) and (8), and such that $p_0 = p_0$ and $a_0 = a_0$.⁴

Having put the existence and uniqueness problems out of the way, we can now proceed to the substantial part of the section, namely the characterization of equilibrium paths. The issue becomes slightly more technical than usual, because we are dealing with a system of three differential equations. However, as we will try to persuade the reader in what follows, the analysis is still quite elementary once some simple (but perhaps relatively unfamiliar) mathematical facts are put into use.

Let the characteristic roots of matrix B be denoted by λ_i , $i = 1, 2, 3$; then, if the λ 's are pairwise distinct, any solution of (14) can be expressed as follows [see, for example, Pontryagin (1962)]:

$$x_t = \sum_{i=1}^3 \omega_i h_i e^{\lambda_i t}, \quad (17)$$

⁴Existence requires, in addition, something like the 'Constrained-Linear-Independence Condition' discussed in Calvo (1985b); in fact, (19b), below, can be used to show that the condition holds in our model when characteristic roots are pairwise distinct.

where $x=(v, p, a)$, ω_i , $i=1, 2, 3$, are arbitrary scalars, and h_i is an eigenvector associated with root λ_i (i.e. $Bh_i = \lambda_i h_i$). By previous considerations, matrix B has only one root with non-negative real parts. Let that root be denoted by λ_3 , it is clear from (17) that a necessary condition for convergence of x is that $\omega_3=0$. Therefore, the other two roots (the ones having the negative real part) will dictate the behavior of *any* equilibrium path (of the linear approximation).

In order to illustrate a case where the overshooting phenomenon can be easily exhibited, let us study the situation where λ_1 and λ_2 are complex conjugates (conditions for the existence of this case will be discussed below). Clearly, then, all the components of x will permanently fluctuate around their steady-state values, so any path starting outside the steady state will necessarily portray overshooting. In fact, more information about the dynamic features of the solution can be obtained if we plot the $\dot{a}=0$ line in the plane (a, p) as in fig. 1. By (8) and (9d), $\delta\dot{a}/\delta a = -z_a < 0$, so we can determine the direction of movement of a in each one of the two zones divided by the $\dot{a}=0$ line. A structural change that requires an appreciation of the real exchange rate only (like in fig. 1), will, therefore, involve an improvement of the trade balance (or balance of payments in the present context), and a later deterioration before the symptoms of overshooting are noticed (i.e. before $p > 0 =$ the steady state of p); the first signs of overshooting will come while the trade balance is already in the red.

We see, therefore, that a lot can be said with very little. To gain more insight, however, some further mathematical elaboration will be of great use. We will now take a closer look at the characteristic polynomial associated with matrix B , i.e.

$$\Psi(\lambda) \equiv \text{determinant of } (B - I\lambda) = \delta^2 \beta f_a z_p - (z_a + \lambda)(\lambda^2 + \delta^2 \beta f_p), \quad (18)$$

where I is the identity matrix.

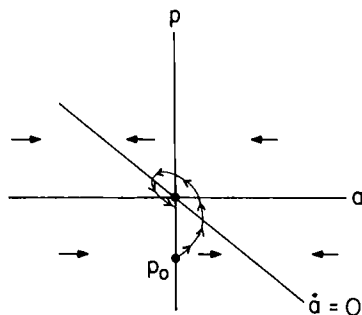


Fig. 1. Equilibrium path: Complex roots.

By (18) we see that by increasing $(f_a z_p)$ we shift up the graph of Ψ in a parallel fashion; thus, given that $\Psi(\lambda) \rightarrow \infty$ as $\lambda \rightarrow -\infty$, there exists some critical number such that, for values of $(f_a z_p)$ larger than that critical number, the roots with negative real parts become complex numbers. This proves that the complex-roots case discussed before is not vacuous. The intuition for this case, and for all the others discussed in this paper, is much harder to come by. The formal analysis has shown us that when the asset effect on the excess demand for home goods and/or the price effect on the excess demand for tradables become sufficiently large, the system will overshoot; to make this fully intuitive *has to be very hard* because we have to remember that our equilibrium concept implies ‘solving out’ for the equilibrium value of v – we are not dealing with a standard 2×2 equation system.

Still some intuitive insight is possible by examining extreme cases. For the sake of brevity, let us consider the case where the price effect in the tradables market becomes unboundedly large (i.e. z_p tends to ∞). Let us assume that the initial conditions are like the ones depicted in fig. 1; a very large z_p would imply a very significant initial reduction in the demand for tradables compared to the long-run equilibrium situation, generating a sizable trade balance and, hence, a high rate of assets accumulation; this situation will not be reversed until p is actually very close to its long-run equilibrium value because z_p is assumed to be very large. In the meantime, p will be moving towards its long-run value because of the excess demand generated by the higher assets and the less-than-long-run-equilibrium value of p . So we have a situation where the level of assets tends to move away from equilibrium much of the time while the real exchange rate is pulled towards steady-state equilibrium. It is, therefore, plausible to expect that p will reach its long-run equilibrium value before a does, at which point in time the upward pressures on p will still exist because the level of assets is higher than that which would generate a zero excess demand, thus giving rise to overshooting.

The above argument suggests a more general reason for overshooting, namely different speeds of adjustment of the asset accumulation and price-change equations; thus, it suggests that one should also get overshooting, given the initial conditions of fig. 1, when the excess demand for home goods becomes very sensitive with respect to the real exchange rate, or when the frequency of price signals increases without bound, or when the sensitivity of the newly-posted price, v , with respect to excess demand, β , increases beyond a certain limit. The conjecture turns out to be correct, but to prove it we need to study the (mathematically much more interesting) case of real negative roots. The following result, proven in the appendix, shows why this is so:

Proposition 1. Given all the other parameters of the model, there exists a critical value of f_p , denoted by $(f_p)^c$, such that if $f_p < (f_p)^c$, then all characteristic

roots are real. Similarly, given all the other parameters of the model, there exists a critical value of $(\delta^2\beta)$, denoted by $(\delta^2\beta)^c$, such that if $(\delta^2\beta) > (\delta^2\beta)^c$, then, again, all roots are real.

We will now examine the case of real roots. Again, denoting the negative roots of Ψ by $\lambda_{1,2}$, and letting $h_i = (h_{i1}, h_{i2}, h_{i3})$, we have, recalling (15) and that $Bh_i = \lambda_i h_i$:

$$\delta(h_{i1} - h_{i2}) = \lambda_i h_{i2}, \quad (19a)$$

$$-z_p h_{i2} - z_a h_{i3} = \lambda_i h_{i3} \quad (19b)$$

[remember that, by definition, $h_i \neq (0, 0, 0)$]. By (18),

$$\lambda_i + z_a \neq 0. \quad (20)$$

Hence, by (19) and (20), all the components of h_i , $i = 1, 2$, are nonzero.

Another straightforward mathematical fact is that along equilibrium solutions that are obtained by setting $\omega_i = 0$ for some $i = 1, 2$ – recall (17) and that our equilibrium concept calls for setting $\omega_3 = 0$ – we have (outside the steady state),

$$p/a = h_{j2}/h_{j3}, \quad j \neq i, \quad j = 1, 2 \quad (21)$$

(the ratio is well defined because we already established that all the components of h are nonzero). By (19b),

$$h_{i2}/h_{i3} = -(\lambda_i + z_a)/z_p. \quad (22)$$

Thus, the slope of the (a, p) path depends on the sign of $(\lambda_i + z_a)$, which is still unknown to us.

The above-mentioned special paths are going to be very useful for characterizing the existence of overshooting. We are going to be particularly interested in (p/a) when λ_i is the maximum of the two negative roots; such a path is called ‘the dominant eigenvector ray’ in figs. 2(a, b). Let $\lambda_2 > \lambda_1$, then the following result is easy to establish (see the appendix):

Proposition 2. If $(p_0/a_0) \neq h_{12}/h_{13}$, then

$$\lim_{t \rightarrow \infty} (p_t/a_t) = h_{22}/h_{23}. \quad (23)$$

Otherwise, $p_t/a_t \equiv p_0/a_0$.

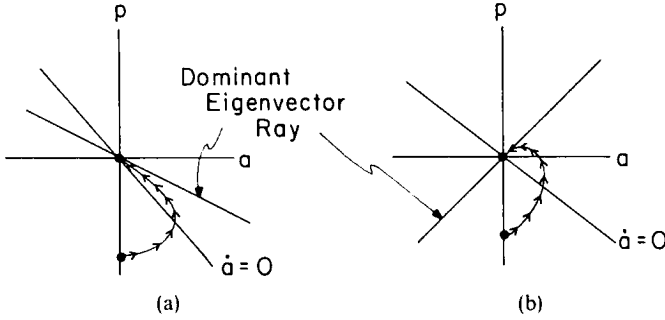


Fig. 2. Equilibrium path: Real roots. (a) Negatively sloped dominant eigenvector ray; (b) positively sloped dominant eigenvector ray.

In words, the above proposition implies that if the initial values of p and a are such that they are not on the ray corresponding to the solutions that only employ the minimum root (by convention, λ_1), then the ratio of p to a will converge to the slope of the dominant eigenvector ray. The application of this mathematical fact to the case at hand is shown in figs. 2(a,b); in fig. 2(a) the slope of the dominant eigenvector ray is negative, and thus when the initial conditions are of the type that we have been discussing all along, the path has to necessarily behave like depicted by the arrowed curve. The equilibrium path cannot cross the dominant eigenvector ray, because the latter is a solution to the differential equation system; hence, if the arrowed curve were to touch it, the economy would indefinitely stay on it. Consequently, overshooting will not be generated. The situation varies radically, however, if the dominant eigenvector ray were to have a positive slope; we depict this in fig. 2(b), where, by Proposition 2, we can infer that the equilibrium path (again, denoted by an arrowed curve) has to cross the horizontal line, implying overshooting. The next proposition (proven in the appendix), completely characterizes these two cases:

Proposition 3. The dominant eigenvector ray has a positive slope if

$$\delta(-\beta f_p')^{\frac{1}{2}} > z_a. \quad (24)$$

On the other hand, if the inequality in (24) is reversed, then the dominant eigenvector ray has a negative slope.

An important implication of the above discussion is that, as conjectured, after a structural change of the type we have been discussing so far (i.e. a change in the steady-state real exchange rate, with no modification in the long-run equilibrium value of assets), overshooting is more likely to happen the higher is the frequency of the price-change signal (i.e. the higher is δ) –

or, equivalently, the smaller the average length of price quotations, $(1/\delta)$ – and the higher is the sensitivity of the newly-posted price, v , with respect to home-goods market conditions, β , and to relative prices, $(-f_p)$.

A better understanding of the importance of the asset effect in the tradables market in relation to the overshooting phenomenon is obtained by examining the case where $z_a = 0^5$ – where, according to Proposition 3 and previous discussion, overshooting would necessarily be a feature of the equilibrium solution if the initial conditions are like the ones depicted in fig. 2. Since the initial level of p is below its long-run value (while $a_0 = 0$), then there exists an excess demand for home goods, and an excess supply of tradable goods (trade balance surplus); the latter implies that assets will be accumulated, bringing further demand pressure for nontradables. This will result in a rising p (an appreciation of the real exchange rate) that will tend to deteriorate the balance of trade relative to the first stages of the equilibrium path; but as long as $p < 0$ (long-run level of p), the trade balance will be in surplus because of the following reasons: (1) the trade balance is independent of the level of assets, hence there is only one value of p that sets it in equilibrium (by construction that level of p has to be zero), and (2) the trade balance is, by (9c) a decreasing function of p . Consequently, a cannot go back to its long-run equilibrium ($=0$) unless p becomes positive, i.e. unless the real exchange rate appreciates over its long-run level: overshooting.

The present economy satisfies a basic property that is worth emphasizing. As shown above, the case for overshooting is stronger the smaller is the ‘life’ of price quotations, i.e. the closer the economy is to a situation of fully flexible prices; interestingly, however, as argued at the end of section 2, overshooting would never occur if prices were actually fully flexible. We have found, therefore, that there is a *fundamental* qualitative difference between the predictions of a staggered-prices model and a model with full price flexibility.⁶

4. Extensions

The model is easily extended to incorporate interest-rate effects. In a perfect foresight context the latter do not add new differential equations if the relevant real rates of interest involve the price of tradables and the average price level for nontradables (p in our notation), see Calvo (1982a, 1982c). Along these lines one can show that if interest rates have a negative effect on aggregate spending, the probability of overshooting after a struc-

⁵This special case is discussed only to make Proposition 3 more intuitive; however, it is clear that the mechanics will not be very different from the case where z_a is ‘small’.

⁶It should be pointed out, however, that, given (9), the quantitative importance of overshooting goes to zero as $\delta \rightarrow \infty$; however, see Calvo (1982c) for an example where this is not so when $f_p > 0$.

tural change is enhanced. The intuition is as follows: consider the initial conditions of fig. 1; in its way to a steady state, the system will experiment inflation of home-goods prices, which will tend to reduce the real interest rate of home goods, and thus expand aggregate demand; so, introducing an interest rate effect, adds a new stimulating factor to aggregate demand, speeding up the adjustment process of home-goods prices, and thus reinforcing the other elements that promote overshooting [for a formal proof, see Calvo (1982c)].

Another extension that is of some interest for semi-industrialized countries is to allow for the possibility of contractionary devaluation [see Diaz-Alejandro (1963) and Krugman and Taylor (1978) for discussion and motivation of this case]. In our simple model this can be formally captured by assuming that (at least, locally) the aggregate demand for nontradables increases with a rise of its relative price, i.e. $f_p > 0$, contrary to (9a). In Calvo (1982a) it is shown that for a given initial condition on p , our model generates a continuum of equilibrium paths with contractionary devaluation if there is no asset effect in the excess demand for nontradables. Interestingly, this is no longer necessarily so in the present model; for, the inequalities in (16) – which, as in section 3, can be used to prove uniqueness and existence – would still hold when $f_p > 0$ given that f_p is less than $(z_p f_a / z_a)$. As shown in Calvo (1982c), those inequalities are satisfied for some plausible choice of parameters. In addition, our simulations suggest that one tends to get more 'realistic' paths when devaluation is contractionary.

Appendix

Proof of Proposition 1. Choose a value of λ (λ^* , say) such that $(\lambda^* + z_a) < 0$; given the other parameters of the model, one can now find f_p negative enough such that $\Psi(\lambda^*) < 0$ [recall (18)]. Let us take any arbitrary value for f_p satisfying the previous inequality and let us call it $(f_p)^c$; clearly, by (18), $\Psi(\lambda^*) < 0$ for all $f_p \leq (f_p)^c$. In addition, by (18), $\Psi(\lambda) \rightarrow \infty$ as $\lambda \rightarrow -\infty$; this implies that if $f_p \leq (f_p)^c$, then the characteristic polynomial exhibits at least one negative real root; but since this polynomial has a positive real root, it follows that all the roots are real numbers, proving the first part of the proposition. The second part is shown in a similar manner. Q.E.D.

Proof of Proposition 2. By (17), and the fact that equilibrium paths have $\omega_3 = 0$, we have

$$\frac{p_t}{a_t} = \frac{\omega_1 h_{12} e^{-(\lambda_2 - \lambda_1)t} + \omega_2 h_{22}}{\omega_1 h_{13} e^{-(\lambda_2 - \lambda_1)t} + \omega_2 h_{23}}. \quad (\text{A1})$$

The condition for the first part of the proposition is equivalent to saying that

$\omega_2 \neq 0$; hence, (A1) implies (23) given the assumption that $\lambda_2 > \lambda_1$. The last part of the theorem is trivial given (A1) because it corresponds to the case where $\omega_2 = 0$. Q.E.D.

Proof of Proposition 3. Consider the polynomial

$$\Psi_0(\lambda) = \Psi(\lambda) - \delta^2 \beta f_a z_p. \quad (\text{A2})$$

By (18), it is easy to establish that the roots of Ψ_0 are: $-z_a$ and $\pm \delta(-\beta f_p)^{\frac{1}{2}}$ and, furthermore, that for $i = 1, 2$,

$$\min\{-z_a, -\delta(-\beta f_p)^{\frac{1}{2}}\} < \lambda_i < \max\{-z_a, -\delta(-\beta f_p)^{\frac{1}{2}}\}. \quad (\text{A3})$$

Hence, if (24) holds, then $\lambda_i < -z_a$; therefore, recalling (9), the r.h.s. of (22) is positive for $i = 1, 2$, implying that the dominant eigenvector ray is positive as stated in the proposition. The converse is shown in a similar manner. Q.E.D.

References

- Calvo, G.A., 1977, The stability of models of money and perfect foresight, *Econometrica* 45, Oct., 1737–1739.
- Calvo, G.A., 1982a, Staggered contracts and exchange rate policy, Discussion paper series no. 129, Columbia University; also in: J.A. Frenkel, ed., *Exchange rates and international economics* (University of Chicago Press, Chicago, IL).
- Calvo, G.A., 1982b, Staggered contracts in a utility-maximizing framework, Discussion paper series no. 130, Columbia University; also in: *Journal of Monetary Economics*, Sept., 383–398.
- Calvo, G.A., 1982c, Real exchange rate dynamics with fixed nominal parities: On the economics of overshooting and interest rate management with rational price setting, Discussion paper series no. 162 (Columbia University, August).
- Calvo, G.A., 1983, Trying to stabilize: Some theoretical reflections based on the case of Argentina, Discussion paper series no. 128, Columbia University; also in: P. Aspe et al., eds., *Financial policies and world capital markets: The problem of Latin American countries* (University of Chicago Press, Chicago, IL).
- Calvo, G.A., 1985a, Fractured liberalism: Argentina under Martinez de Hoz, Discussion paper series no. 275, Columbia University; also in: *Economic Development and Cultural Change* 34, April 1986, 511–533.
- Calvo, G.A., 1985b, Reserves and the managed float: A search for the essentials, *Journal of International Money and Finance* 4, March, 43–60.
- Calvo, G.A. and C.A. Rodriguez, 1977, A model of exchange rate determination under currency substitution and rational expectations, *Journal of Political Economy* 85, April, 617–625.
- Diaz-Alejandro, C.A., 1963, A note on the impact of devaluation and the redistribution effect, *Journal of Political Economy* 71, 577–580.
- Diaz-Alejandro, C.A., 1981, Southern cone stabilization plans, in: ch. 4 of W. Cline and S. Weintraub, eds., *Economic Stabilization in Developing Countries* (Brookings Institution, Washington, D.C.).
- Dornbusch, R., 1982, Stabilization policies in developing countries: What have we learned?, *World Development* 10, Sept., 141–153.
- Edwards, S., 1985, Stabilization with liberalization: An evaluation of ten years of Chile's experiment with free market policies, *Economic Development and Cultural Change* 33, Jan., 223–254.

- Harberger, A.C., 1982, The Chilean economy in the 1970's: Crisis, stabilization, liberalization, reform, Carnegie-Rochester Conference Series on Public Policy 17, Autumn, 115–152.
- Kouri, P.J.K., 1976, The exchange rate and the balance of payments: A monetary approach, *Scandinavian Journal of Economics* 78, no. 2, 280–304.
- Krugman, P. and L. Taylor, 1978, Contractionary effects of devaluation, *Journal of International Economics* 8, 445–456.
- Obstfeld, M., 1984, Capital flows, the current account, and the real exchange rate: Consequences of stabilization and liberalization, Mimeo, Dec.
- Pontryagin, L.S., 1962, *Ordinary differential equations* (Addison-Wesley Pub. Co. Inc., New York).
- Rodriguez, C.A., 1982, The Argentine stabilization plan of December 20th, *World Development* 10, Aug., 226–238.
- Rodriguez, C.A. and L.A. Sjaastad, 1979, El atraso cambiario en Argentina: Mito o realidad?, C.E.M.A., Documento de Trabajo No. 2, June.
- Sargent, T. and N. Wallace, 1973, The stability of models of money and growth with perfect foresight, *Econometrica* 41, 1043–1048.